

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Annexure A

Total Marks: 25

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

	Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
A	Understanding of Concepts	15% 1.5	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
B	Comparison Depth	20% 2	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
C	Criteria Selection	10% 1	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
D	Analytical Skills	15% 1.5	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
E	Organization & Structure	10% 1	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
F	Use of Examples / Evidence	10% 1	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
G	Clarity of Presentation	10% 1	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
H	Conclusion & Insights	10% 1	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

	A	B	C	D	E	F	G	H	
Q1	1	2	1	1	1	1	1	0.5	8.5
Q2	1	2	1	1	1	1	1	0.5	8.5
Q3	1.5	1	0.5	1.5	0.5	1	0.5	0.5	7
Q4	1.5	1	0.5	1.5	0.5	1	1	0.5	7.5
Q5	1	1.5	1	1	0.5	1	1	0.5	7.5

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1.Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

Answers:

Comparative Analysis: Star Topology vs Mesh Topology

Feature	Star Topology	Mesh Topology
Maintenance	Easy to maintain	Difficult to maintain
Troubleshooting	Faults can be easily identified	Troubleshooting is complex
Connections	Each device connects to a central switch/hub	Devices have multiple connections
Maintenance Cost	Low to moderate	High
Reliability	Good	Very high
Complexity	Simple structure	Highly complex structure
Time Required	Less time for maintenance	More time required

2.Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

Answers:

Comparative Analysis: Mesh Topology vs Star Topology

Feature	Mesh Topology	Star Topology
Fault Tolerance	Very high because multiple paths are available	Moderate because devices depend on the central switch/hub
Reliability	Highly reliable; one link failure usually does not stop communication	Reliable for individual link failures, but central device failure affects the network
Data Paths	Multiple paths between nodes	Mainly through the central device
Failure Impact	Failure of one link has limited effect	Failure of the central switch/hub can affect the entire network
Maintenance	Complex and difficult	Easier to maintain and troubleshoot
Cost	Very expensive due to large cabling requirements	Less expensive compared to mesh
Scalability	Difficult to expand because many connections are required	Easy to add new devices
Performance	Excellent due to dedicated links and alternate paths	Good performance, depending on the central device
Suitable For	Critical and high-reliability networks	LANs, offices, schools, and general networks

3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

Answers:

Comparative Analysis: Bus Topology vs Mesh Topology

Feature	Bus Topology	Mesh Topology
Installation	Simple and easy to install	Complex and time-consuming
Cabling	Uses a single backbone cable	Requires multiple cables between nodes
Cost	Low cost	High cost
Connections	Minimal connections	Each node has multiple connections
Maintenance	Easy for small networks	Difficult due to many connections
Scalability	Limited; adding devices may affect performance	Difficult and expensive as devices increase
Reliability	Lower; backbone failure can affect the network	Very high due to multiple alternative paths
Suitable For	Small and simple networks	Critical networks requiring high reliability

4.Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

Answers:

Comparative Analysis: Hybrid Topology vs Star Topology

Feature	Hybrid Topology	Star Topology
Scalability	Very high; different topologies can be combined and expanded	Moderate to high; depends on the central switch/hub
Expansion	New network segments can be added easily	New devices require available ports
Flexibility	Highly flexible because different topologies can be combined	Less flexible because it mainly follows a central-device structure
Cost	Generally higher due to complex design and equipment	Lower compared to hybrid topology
Installation	More complex	Simple and easy
Maintenance	More difficult because of the mixed structure	Easier to manage and troubleshoot
Central Device Dependency	Depends on the design; not always dependent on one device	Highly dependent on the central switch/hub
Suitable For	Large organizations and growing networks	Small to medium-sized LANs

5.In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

Answers:

Comparative Analysis: Star, Mesh, Bus and Hybrid Topologies – Maintenance

Feature	Star Topology	Mesh Topology	Bus Topology	Hybrid Topology
Maintenance	Easy	Difficult	Moderate to difficult	Moderate
Troubleshooting	Easy to identify faulty node or cable	Time-consuming due to many connections	Difficult because backbone faults affect the network	Requires knowledge of different topologies
Network Structure	Simple	Complex	Simple	Complex
Fault Isolation	Easy	Possible but complex	Difficult	Depends on the topology used
Maintenance Cost	Low to moderate	High	Low	Moderate to high
Required Skill	Basic networking skills	Advanced networking skills	Basic to moderate skills	Skilled administration may be required
Failure Impact	Usually affects only one device	Usually limited due to alternate paths	Backbone failure can affect the entire network	Depends on the affected segment